

Utkarsh Ranjan

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EDUCATION

University of California, San Diego, USA
Master of Science in Computer Science and Engineering

Incoming Fall'26
GPA: - / 4.0

Indian Institute of Technology Bombay, India
Bachelor of Technology (Honors) in Computer Science and Engineering

Nov 2020 - May 2024
GPA: 9.05/10.0

RESEARCH INTERNSHIPS & KEY PROJECTS

BharatGen: English Language Model

Guide: Prof. Ganesh Ramakrishnan | B.Tech Project

Spring 2024
IIT Bombay

- Supported training & evaluation of BharatGen, a **7B+** parameter LLM covering **22 Indian languages**.
- Implemented **distributed training** using **Megatron-DeepSpeed** with data, tensor, and pipeline parallelism.
- Benchmarked model inference on **NVIDIA A100** and **Intel Gaudi2** accelerators across **30+** LM evaluation tasks.

Decentralized Learning over Wireless Sensor Network

Guide: Prof. Anupam Chattopadhyay | Research Project

Jan 2022 – July 2022
NTU, Singapore

- Conducted an extensive literature survey on recent developments in decentralized and distributed learning algorithms.
- Designed and built a client-server communication simulator to evaluate performance of multiple algorithms.
- Implemented adaptive neighbor matching, improving convergence over a complete graph by **1.25x**.

DeepSynth: ML framework for Grammar-Based Program Synthesis

Guide: Nathanaël Fijalkow, CNRS | Research Internship

Summer 2022
LaBRI, Bordeaux, France

- Enhanced the algorithm used for enumerating programs generated using probabilistic context-free grammar (**PCFG**).
- Introduced discrete probability tuples, reducing multiplication computations and achieving **93.1%** accuracy.

RDF to Natural Language Conversion

Prof. Pushpak Bhattacharyya | R&D Project

Spring 2023
LG Soft India - IIT Bombay

- Finetuned **T5-base** on **A100 (80GB) GPU** for converting knowledge graphs in RDF (triples) format into text.
- Achieved **SOTA** performance of **58.9%** BLEU, **0.69** BLEURT and **0.43** METEOR scores on WebNLG'17 corpus.

Static Analysis for Optimized Garbage Collection

Guide: Prof. Swaprava Nath | Course Project: Code Optimization for OO Language

Spring 2024
IIT Bombay

- Performed inter-procedural analysis on Java code to detect when objects escapes for early garbage collection.
- Transformed bytecode and measured **2x** performance improvements in the OpenJ9 JVM interpreter-only mode.

C like Compiler

Guide: Prof. Amitabha Sanyal | Course Project: Compilers

Spring 2023
IIT Bombay

- Developed a compiler for a subset of C, supporting **functions**, **scope levels**, **recursion** and **control sequences**.

WORK EXPERIENCE

Stripe: Software Engineer, Product

Full-Time Software Engineering

June 2024 - Present
Stripe, Bangalore, India

- Working in the **Financial Reporting** Team serving **2M** merchants, including **150K** Monthly Active Users.
- Spearheading **Agentic Contact Support** for financial reporting, aiming to integrate Reporting API into Stripe Assistant, projected to resolve **2,600** monthly support tickets and save **\$1.2M** annually.
- Contributed to **Microsoft's** EPA reporting system leveraging **PySpark** and **Airflow** for distributed data processing, integrating transaction data across multiple payment flows, which helped secure the **\$100M contract** over 5-years.

Prodigal (YC S'18): Machine Learning Intern

Machine Learning Intern | DSML Team

Summer 2023
Prodigal, Mumbai, India

- Automated the Right Party Check verification of customers using heuristics and **NLP** techniques on call transcripts.
- Pioneered a **semi-supervised** approach using sentence embeddings and clustering to accurately label call segments.

SCHOLASTIC ACHIEVEMENT

- Selected among the **2000 Top Global CS** students for the **Y Combinator's** invite-only AI Startup School. 2025
- Achieved **All India Rank 356** in JEE-Advanced out of 240 thousand eligible candidates 2020
- Awarded **Gold Medal** for ranking among the **Top 46** Nationwide in the Indian National Chemistry Olympiad. 2020
- Achieved **All India Rank 174** and was awarded KVPY fellowship by IISc Bangalore. 2019

POSITIONS OF RESPONSIBILITY

Institute Student Mentor, Student Mentor Programme, IITB

May 2023 - June 2024

Department Academic Mentor, CSE Dept., IIT Bombay

May 2022 - July 2023

Team Member, IITB Racing Driverless Team

Nov 2020 - April 2021